

EES 102: Introduction to Environmental Sciences

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Image: <https://wiki.seg.org/index.php?curid=13659>

ROCKY BODIES DIFFERENTIATE

Basaltic achondrites



Iron meteorite



Basaltic achondrites and iron meteorites are two sides of the same coin. We find them because some planetesimals that were large enough to melt and differentiate were thereafter broken into pieces during subsequent collisions and those pieces reached Earth.

Rocky bodies differentiate into:

rocky crusts and mantles that are depleted in **siderophiles** and enriched in **lithophiles**.

Metallic cores enriched in **siderophiles** and depleted in **lithophiles**.

**We study the
interior with
seismic waves**

**P(rimary)-
wave
Shadow
Zone**

Compressional waves

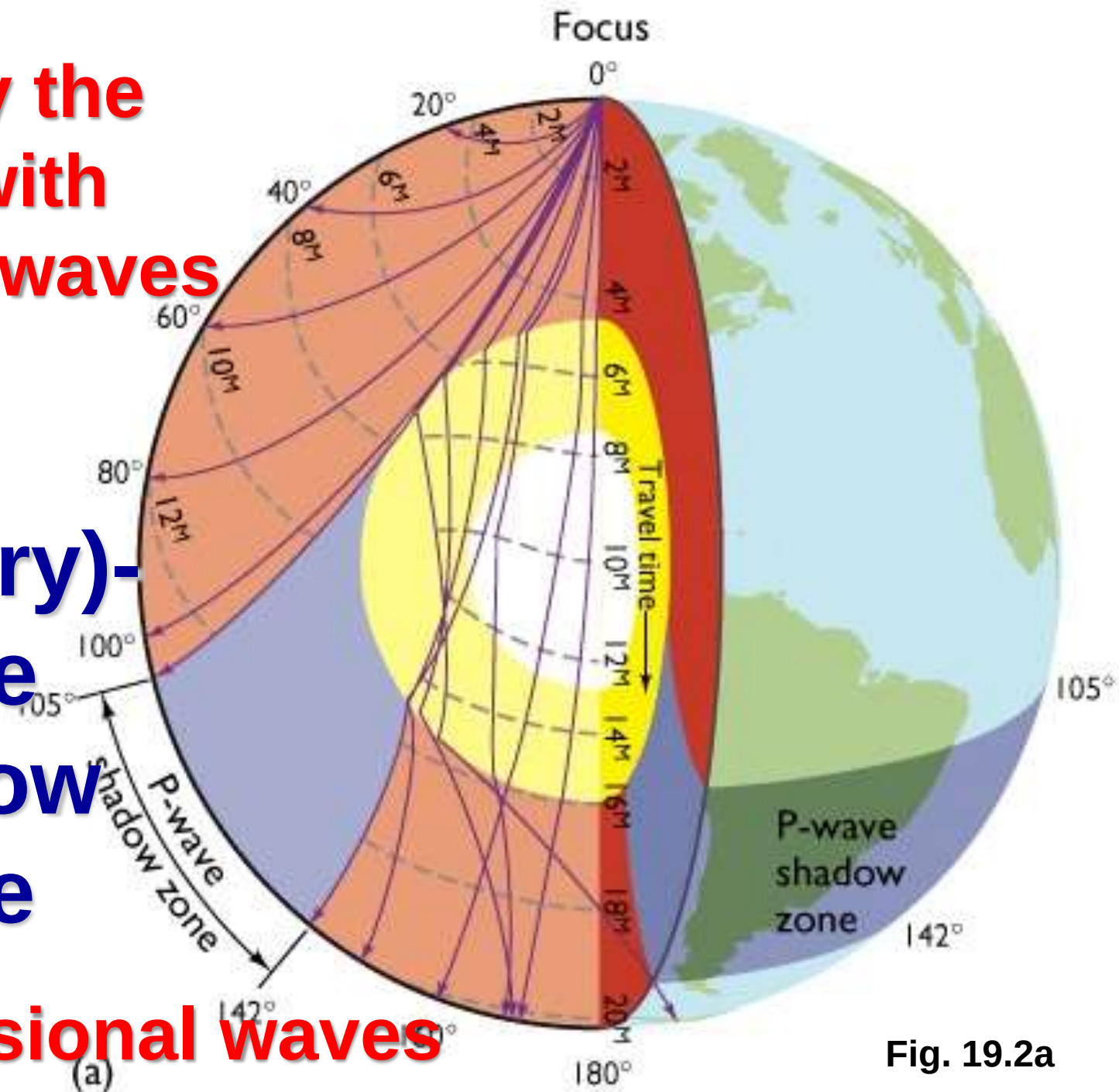


Fig. 19.2a

**We study the
interior with
seismic waves**

**S(ECONDARY)-
WAVE SHADOW
ZONE**

Shear waves

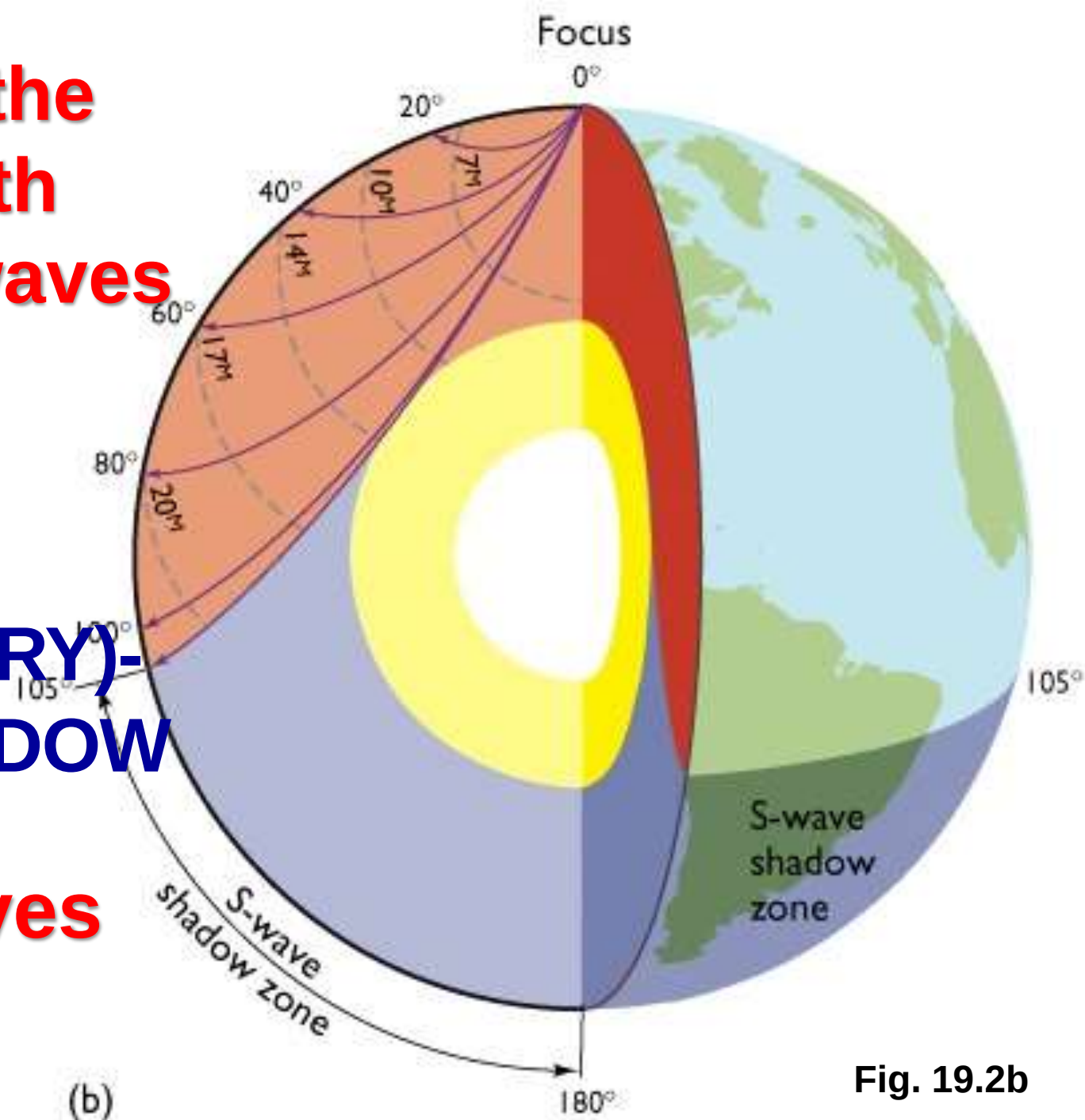


Fig. 19.2b

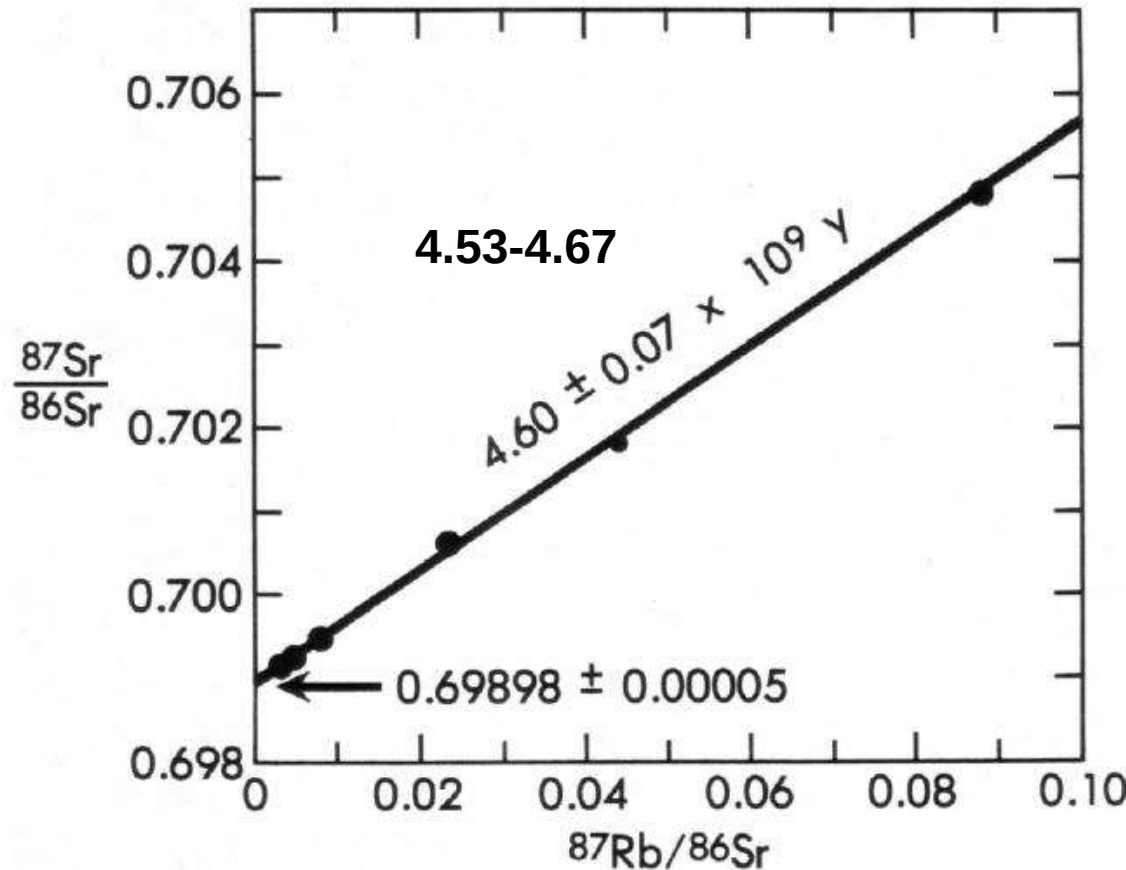
WE DATE THE SOLAR SYSTEM WITH CHONDRITES



- Chondrites are primitive meteorites that represent the original material that aggregated from the dust cloud.
- These have never been heated enough to be melted, come from small bodies and do contain traces in the form of dust grain originating from solar systems other than our own in the form of “pre-solar grains”

The calcium–aluminium-rich inclusion or Ca–Al-rich inclusion (CAI) found in carbonaceous chondrite meteorites. are probably the oldest substances in the Solar System. **The oldest age** was measured in an inclusion of the CV3 carbonaceous chondrite Northwest Africa (NWA) 2364 and **was dated at 4568.22 ± 0.17 Mya using Pb-Pb dating**. They are thought to have formed as fine-grained condensates from a high temperature (>1300 K) gas that existed in the protoplanetary disk .

HOW LONG AGO? RB-SR DATING OF DIFFERENTIATED METEORITES GIVES A DATE ALMOST SAME AS SOLAR SYSTEM AGE



Rb-Sr isochron for the meteorite Juvinas. The points represent analyses on glass, tridymite and quartz, pyroxene, total rock, and plagioclase. After Faure (49). Data from Allegre and others (3).

We date rocks and meteorites using radioactive decay. The take home message here is we know the age of the solar system and Earth from dating meteorites.

CV3 carbonaceous chondrite Northwest Africa (NWA) 2364 and **was dated at 4568.22 ± 0.17 Mya using Pb-Pb dating**

HOW LARGE IS LARGE ENOUGH AND HOW FAST DID MELTING HAPPEN? A PLANETESIMAL CALLED VESTAS HAS THE ANSWER:



The Vestian asteroids and HED meteorites have been broken from Vesta by a collision of a large object

Within 2–3 million years
After the supernova

Within 4–5 million years
After the supernova

Within 6–7 million years
After the supernova

Accretion completed

Complete or almost complete melting due to radioactive decay of ^{26}Al , leading to separation of the metal core

Progressive crystallization of a convecting molten mantle. Convection stopped when about 80% of the material had crystallized

Extrusion of the remaining molten material to form the crust, either as basaltic lavas in progressive eruptions, or possibly forming a short-lived magma ocean.

The deeper layers of the crust crystallize to form plutonic rocks, whereas older basalts are metamorphosed due to the pressure of newer surface layers.

Slow cooling of the interior

THE MOON

Only a little smaller than Mercury (it is like a small planet in two-planet system).

Surface of the moon very different from the surface of Earth yet chemical composition of Moon basalt identical to Earth basalt.

No atmosphere, therefore, no weathering. But the material appears to be the same, chemically and isotopically

FORMATION OF THE MOON



Fig. 1.4

GIANT IMPACT HYPOTHESIS - EXPLAINED

Impact conditions can be found that give rise to a Moon that formed mostly from the mantles of the Earth and impactor, with part of the mantle and the core of the impactor accreting to the Earth, and which satisfy the angular momentum constraints of the Earth–Moon system.

Evidence of remnants of the core of the impactor in the form of seismic anomalies near the Mantle core boundary have recently been detected (Yuan et al. 2023, Nature 623, 95-99).

Some researchers have proposed that the impact may have been a carbon and sulphur rich planetary body from the Mars orbit Grewal et al. 2019

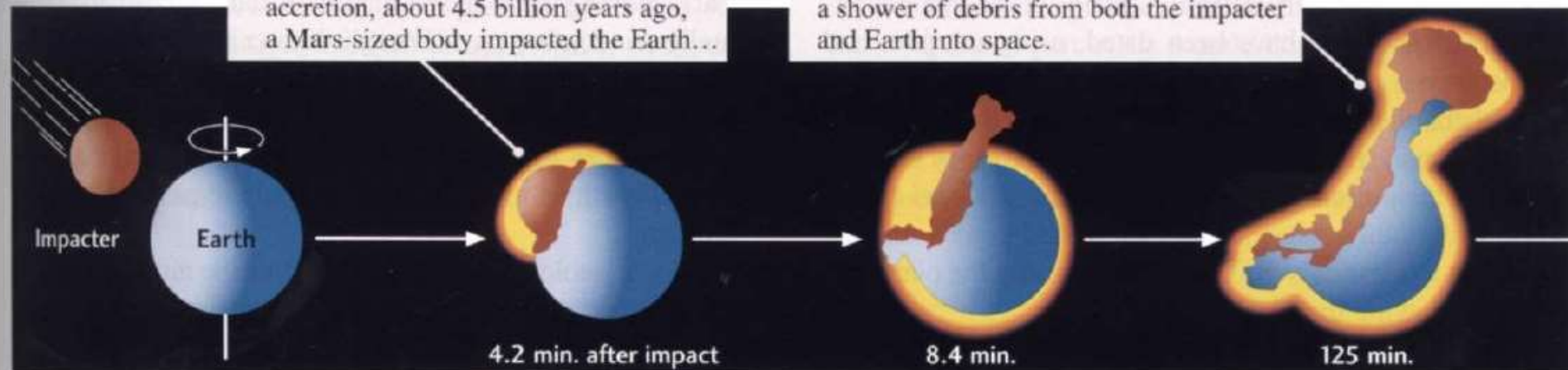
DOI: [10.1126/sciadv.aau3669](https://doi.org/10.1126/sciadv.aau3669).

Recently iron & metal isotopic evidence has been used to argue the impactor may have come from closer to the sun rather than from Mars orbit (Hopp et al. 2025, Science 390, 819-823

Warm silica-rich dust and abundant SiO gas, products of high velocity (> 10 km/s) impacts between rocky bodies, have been detected around the nearby (29 pc distant) young (~12 My old) Beta Pic Moving Group star HD172555 by the Spitzer Space Telescope providing evidence of such impacts

1 During middle to late stages of Earth's accretion, about 4.5 billion years ago, a Mars-sized body impacted the Earth...

2 ...and the giant impact quickly propelled a shower of debris from both the impactor and Earth into space.

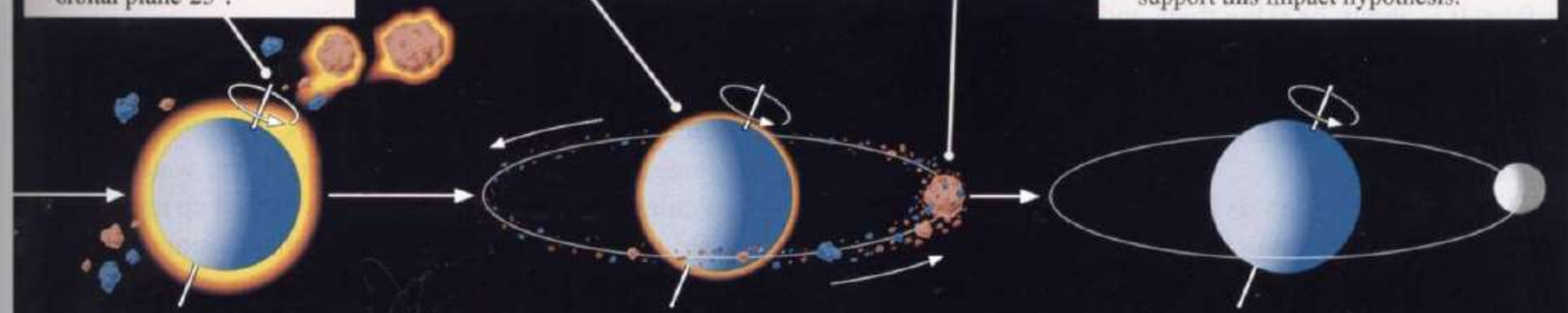


3 The impact sped up Earth's rotation and tilted Earth's orbital plane 23°.

4 Earth re-formed as a largely molten body...

5 ...and the Moon aggregated from the debris.

6 Moon rocks 4.47 billion years old brought back by the *Apollo* astronauts support this impact hypothesis.



GIANT IMPACT HYPOTHESIS - EXPLAINED

Moon formed ~4.47 billion years ago
Based on dating impact shock metamorphism on Vestian meteorites

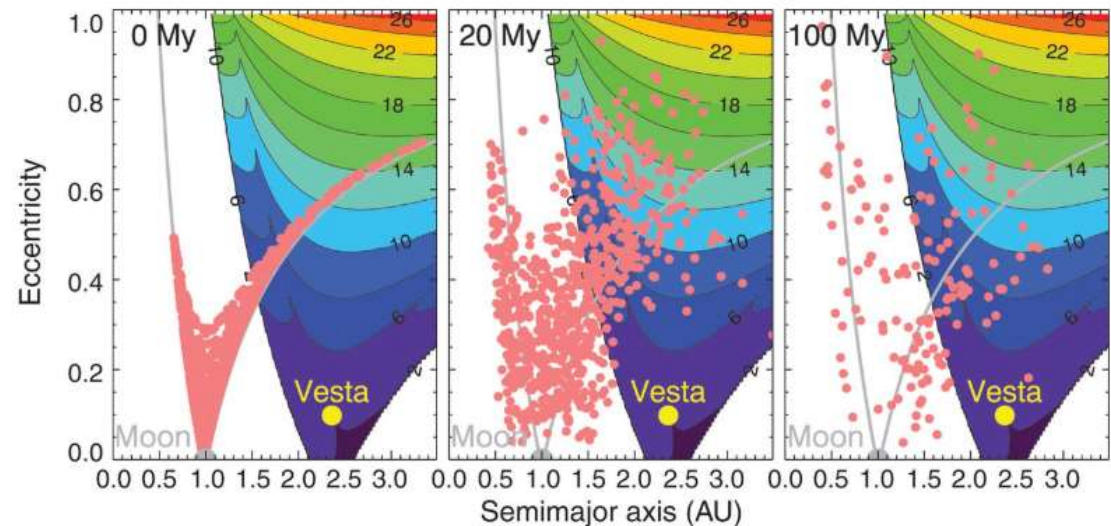
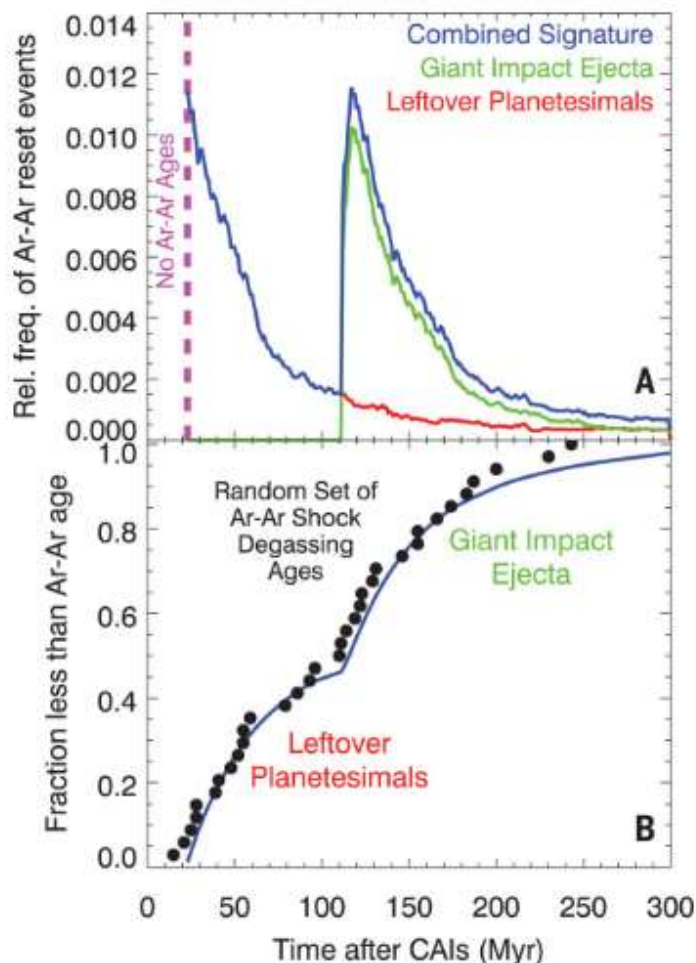
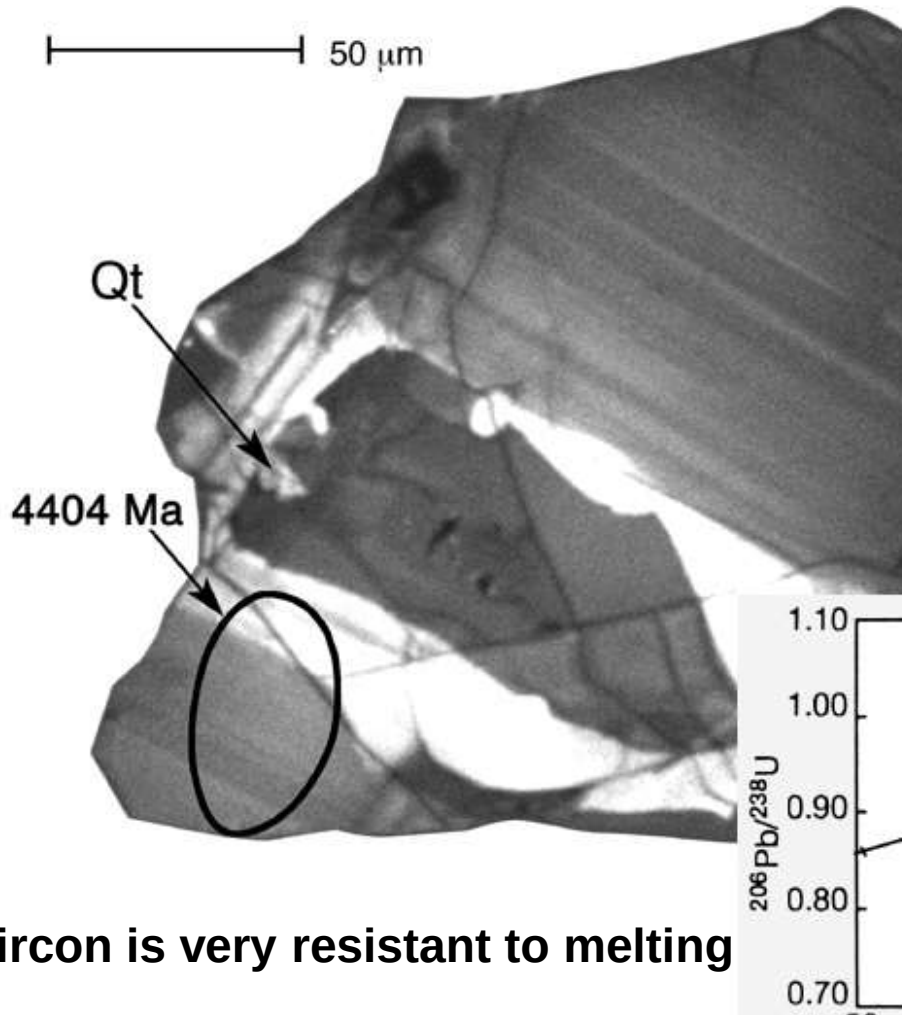


Fig. 3 A sample comparison between our model and randomly derived ^{40}Ar - ^{39}Ar shock degassing ages for asteroidal meteorites.

EARTH

The Earth's oldest mineral grain in a sandstone in Jack Hill is 4.404 Ga old. Since plate tectonics recycles materials the oldest records have not survived



Zircon is very resistant to melting

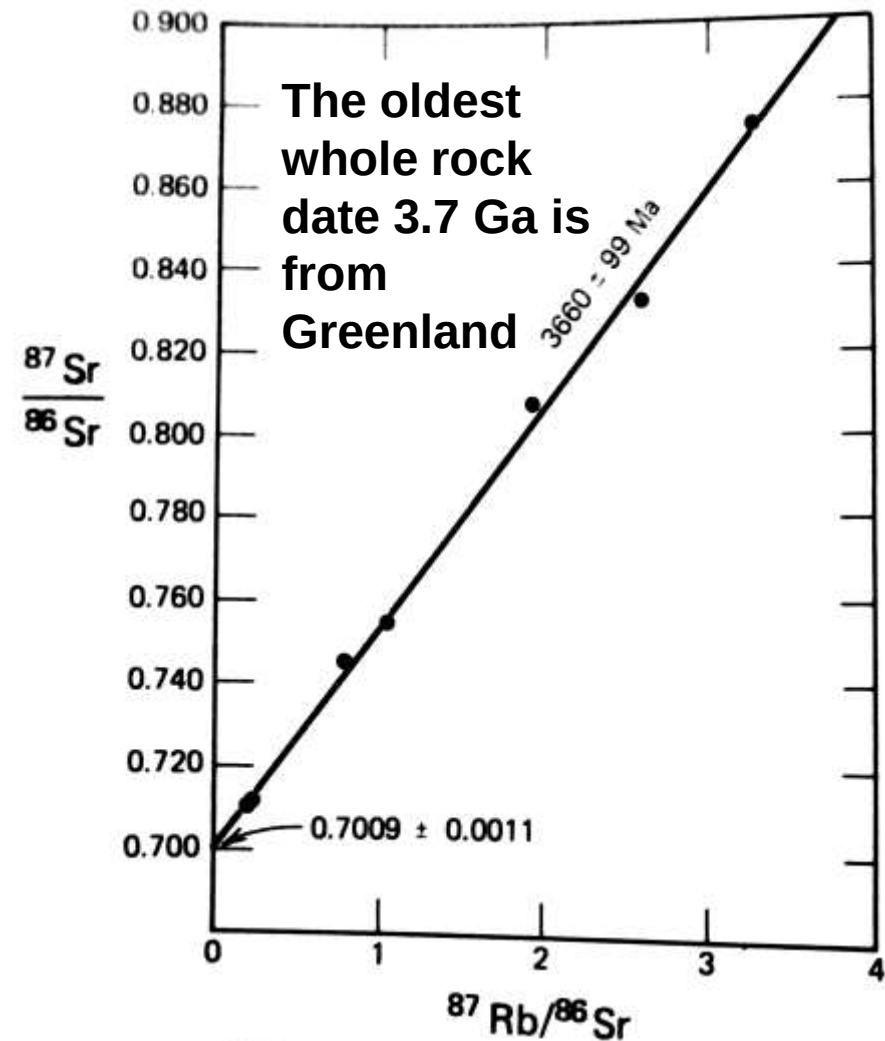


Figure 8.7 Whole-rock isochron for the Amitsoq gneiss from the Quilangarsuit area of the Godthaab district of southwestern Greenland. These are among the oldest terrestrial rocks known. The date probably represents the time elapsed since metamorphism of igneous rocks of granitic composition to form the Amitsoq gneiss. (Data from Moorbath et al., 1972.)

THE EARLY PLANET WAS SUBJECTED TO VIOLENT METEORITE BOMBARDMENT



Craters on moon

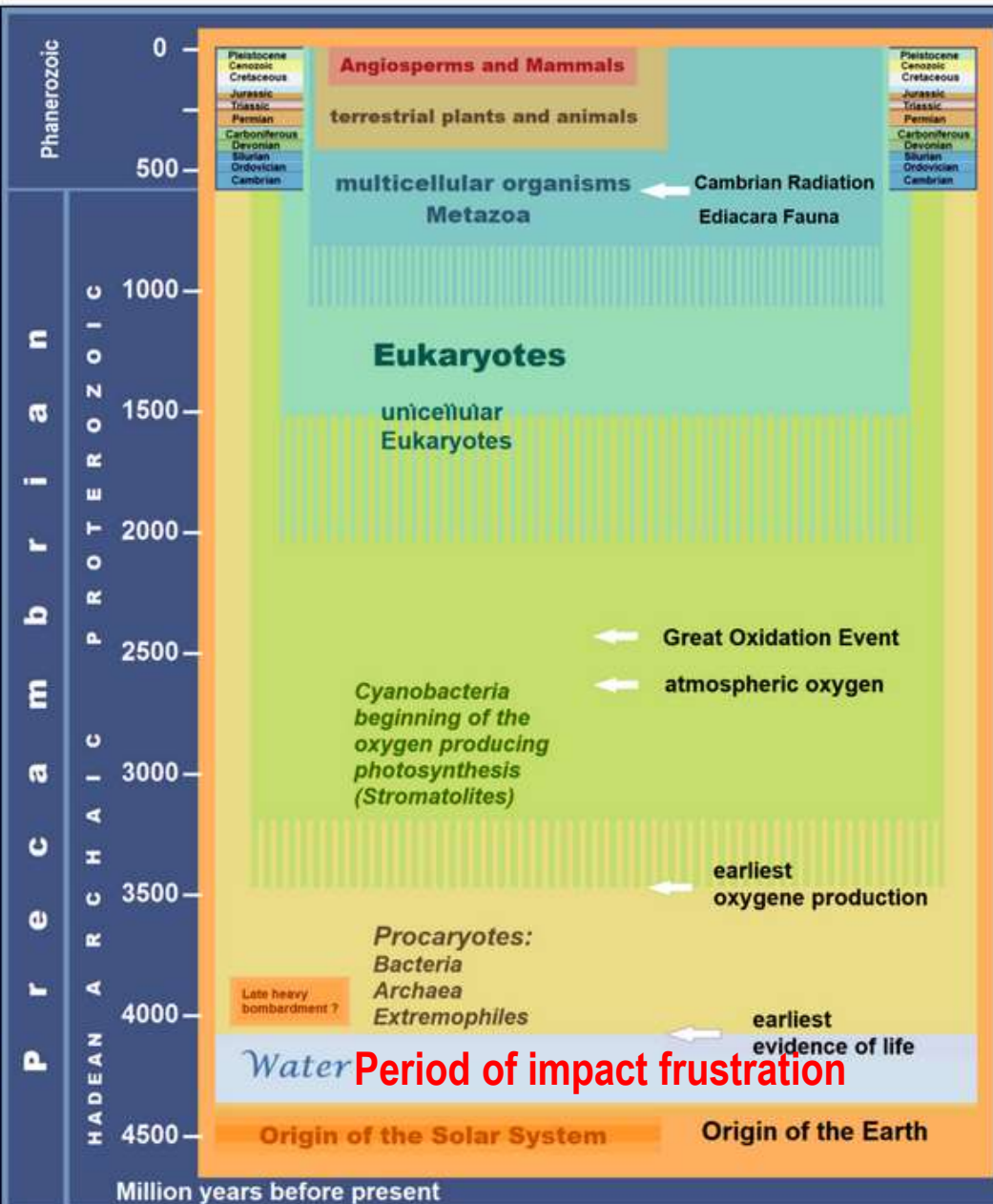
Craters on Earth



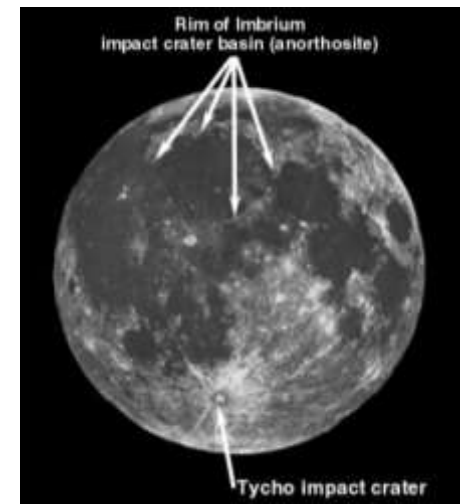
Asteroid belt



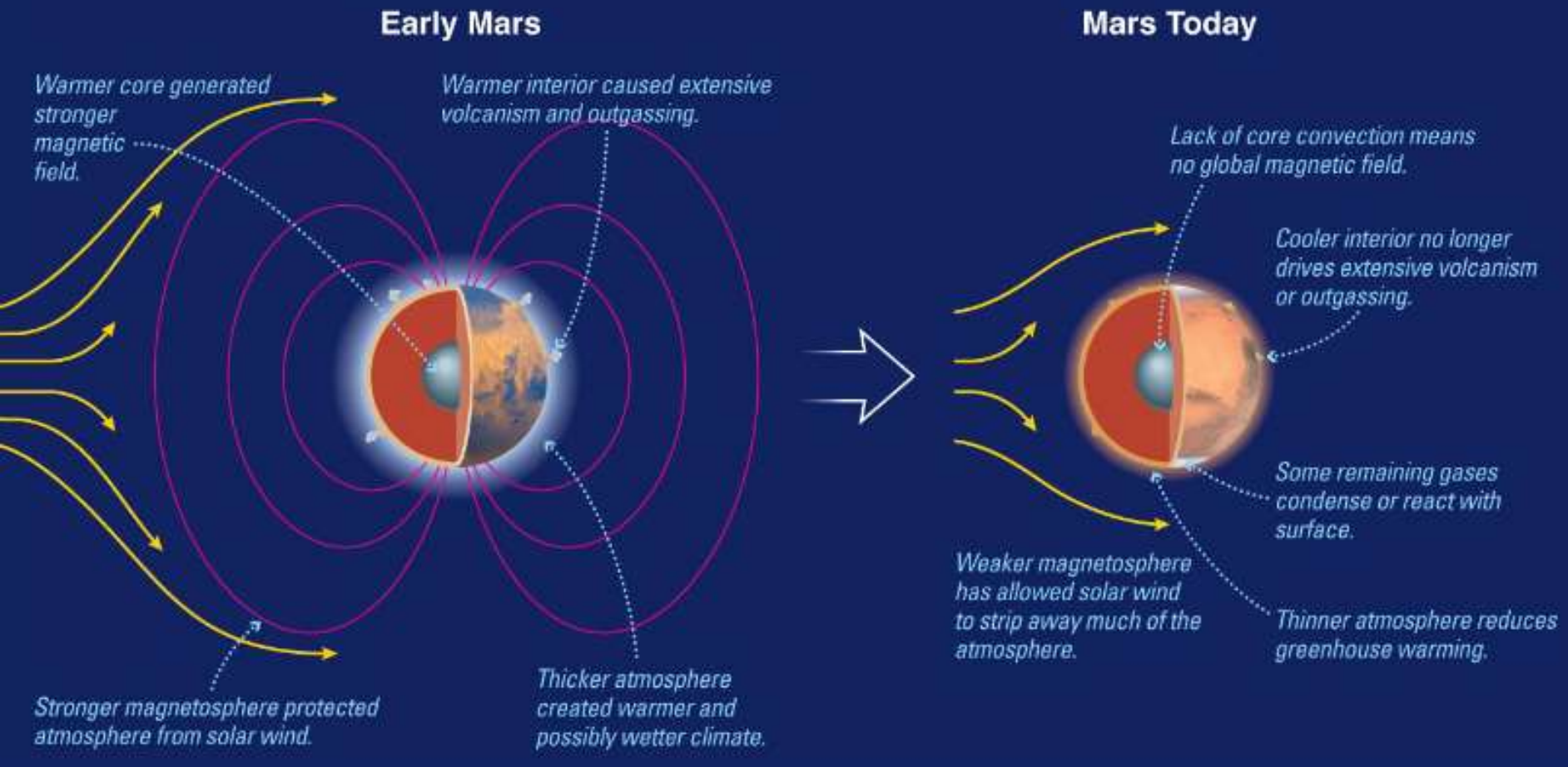
THE HEAVY BOMBARDMENT



- Early Earth had many collisions with impactors of various sizes. However, due to active plate tectonics most preserved impact craters are “small” and young
- Part of the moon surface is quite old while most of the mare basalts are younger than 3.5 Ga and contain only small caters
- Most huge craters on the Moon are on the older surface and their impact-formed rocks have been dated to 4.1 to 3.8 billion years (Ga) indicating that large impactors were more common in the early Hadean
- It may not be a coincidence that the earliest evidence for life dates into the period when the heavy bombardement slowed down



THE ATMOSPHERE



- Planets need an magnetic field to hold onto their atmosphere and protect it from solar winds
- The oldest evidence for the Earths magnetic field is 4.2 Ga old.
- Mars had a magentic field 4 Ga ago and lost it around 3.8 Ga ago.

OUR SECOND ATMOSPHERE CAME OUT OF THE MANTLE

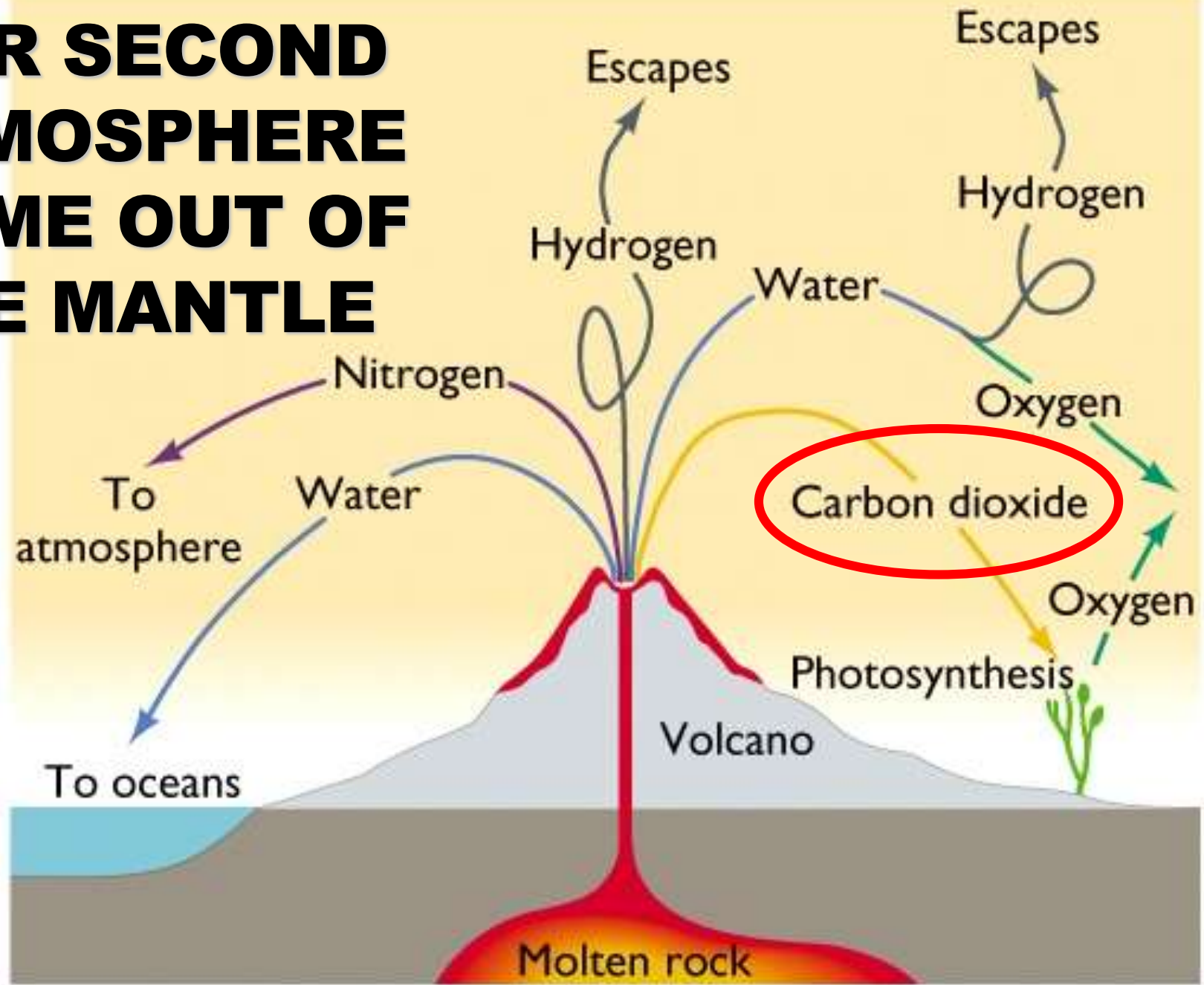


Fig. 1.8

THE SECOND ATMOSPHERE DEGASSED FROM MANTLE MAGMA AND COMPRISED PRIMARILY OF CO₂ WITH SMALL AMOUNTS OF ORGANICS AND WATER WERE DELIVERED BY METEORITES

	VENUS	EARTH	MARS
SURFACE PRESSURE	100,000 mb	1,000 mb	6 mb
COMPOSITION			
CO ₂	>98%	0.03%	96%
N ₂	1%	78%	2.5%
Ar	1%	1%	1.5%
O ₂	0.0%	21%	2.5%
H ₂ O	0.0%	0.1%	0-0.1%

MARS AND VENUS ARE STUCK WITH THEIR 2ND ATMOSPHERE

THIRD ATMOSPHERE OF EARTH EVOLVED THANKS TO PHOTOSYNTHETIC BACTERIA THAT USED CO₂ AND PRODUCED O₂ AND THE MICROBIAL NITROGEN CYCLE

OTHER PLANETS THAT TEACH ABOUT EARLY EARTH

VENUS

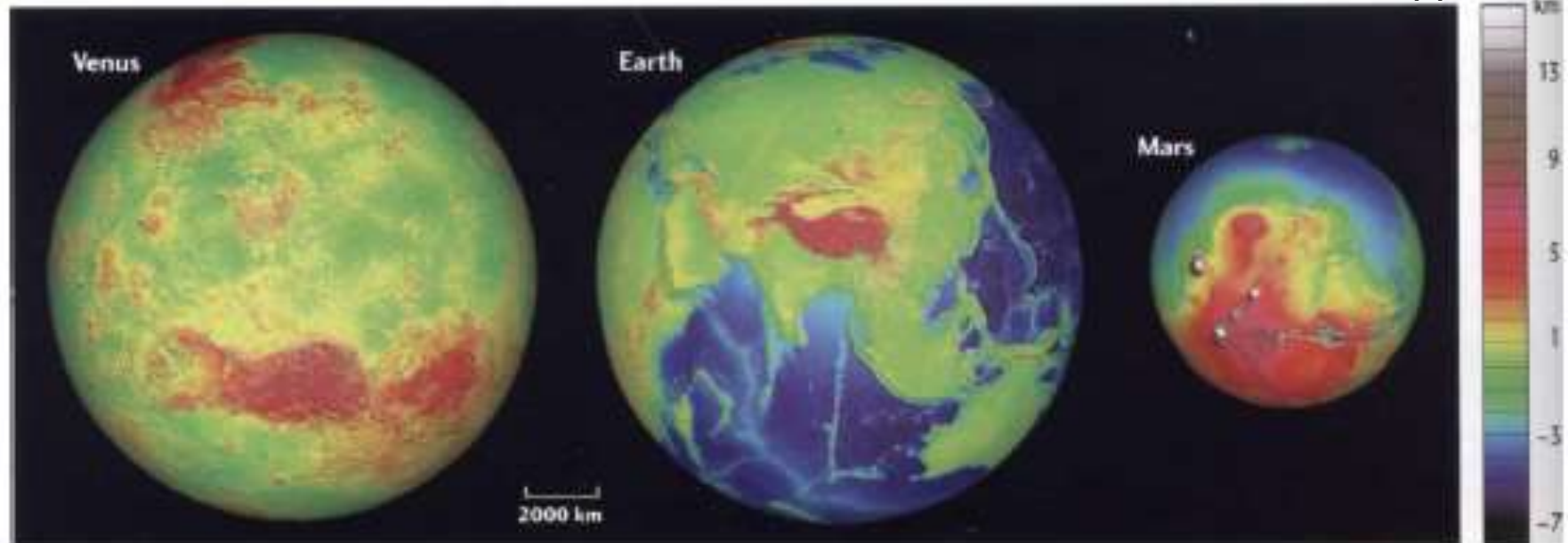
Very hot atmosphere full of GHG. Geologically active with plate tectonics

EARTH

Plate tectonics
Geologically active
Photosynthesis => O₂
rich atmosphere
Nitrogen cycle => N₂

MARS

Used to have an atmosphere & ocean till ~3.5 Ga. Too small to keep a Geodynamo going. Thin atmosphere. Plate tectonics stopped



WE OWE OUR ATMOSPHERE TO THE GEODYNAMO IN OUR CORE

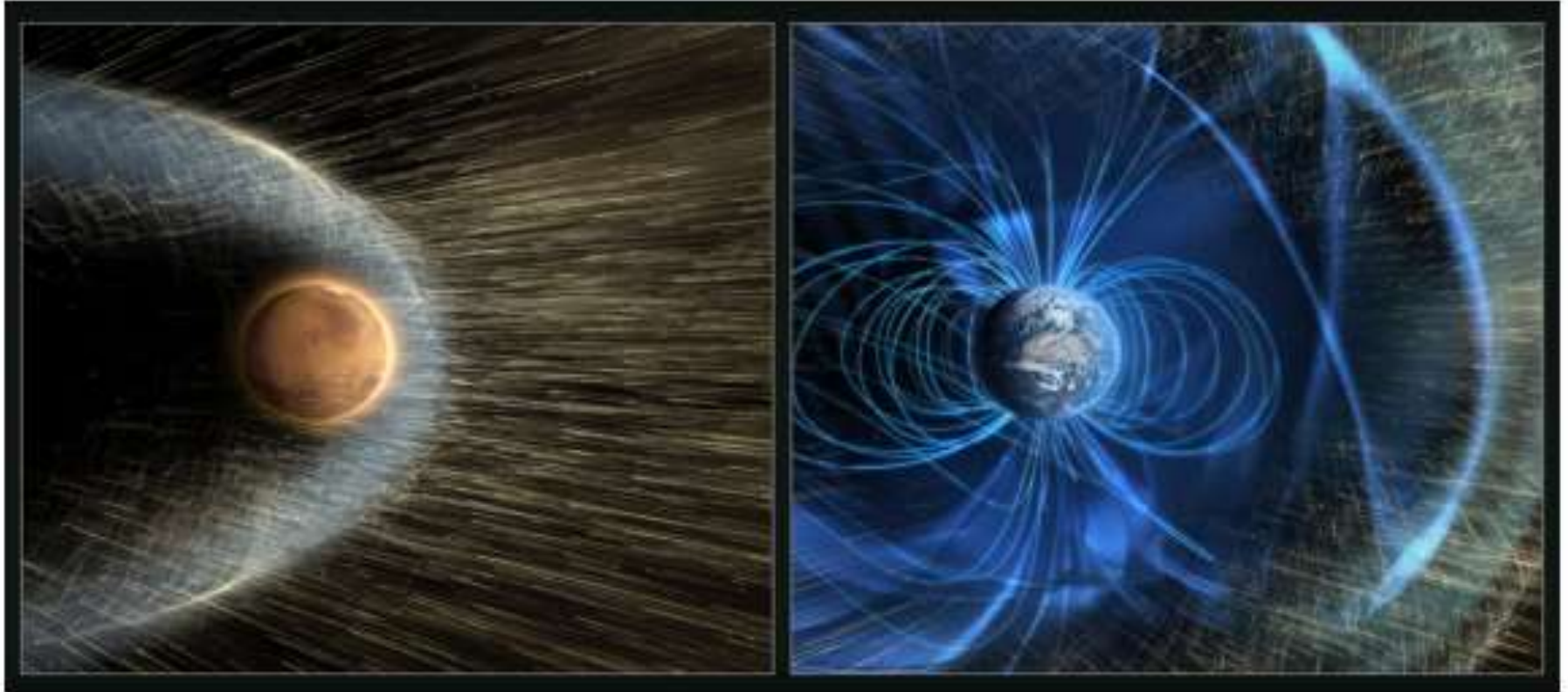
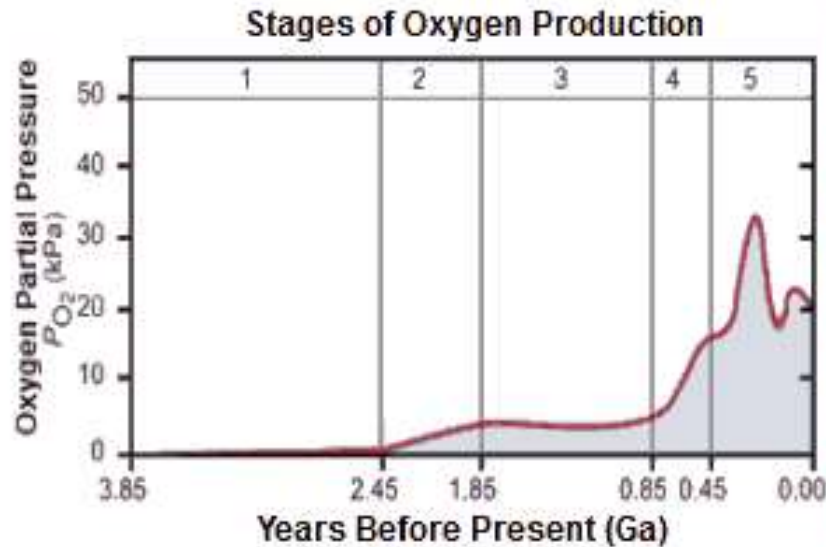


Figure 1. Solar wind attacks Mars and Earth atmosphere.

THE FIRST MASS EXTINCTION WAS TRIGGERED BY OXYGEN



Banded Iron Formation (BIF) from Australia

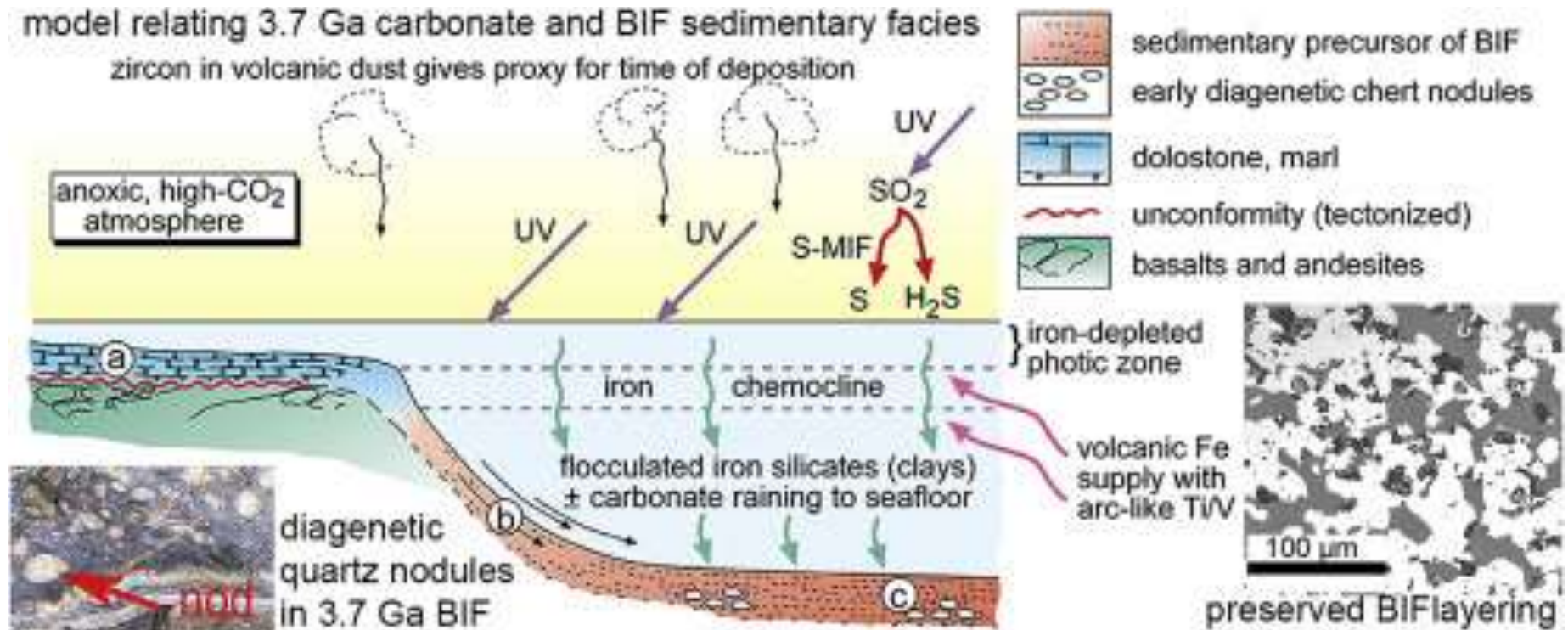
Stromatolites are created by photosynthetic cyanobacteria, the oldest are 3.5 Ga old

The red bands are hematite an iron III mineral

It resulted in the extinction of many forms of chemoautotrophs to whom oxygen is toxic. Some chemoautotrophs still reside in extreme habitats e.g. around deep sea volcanoes as extremophiles.

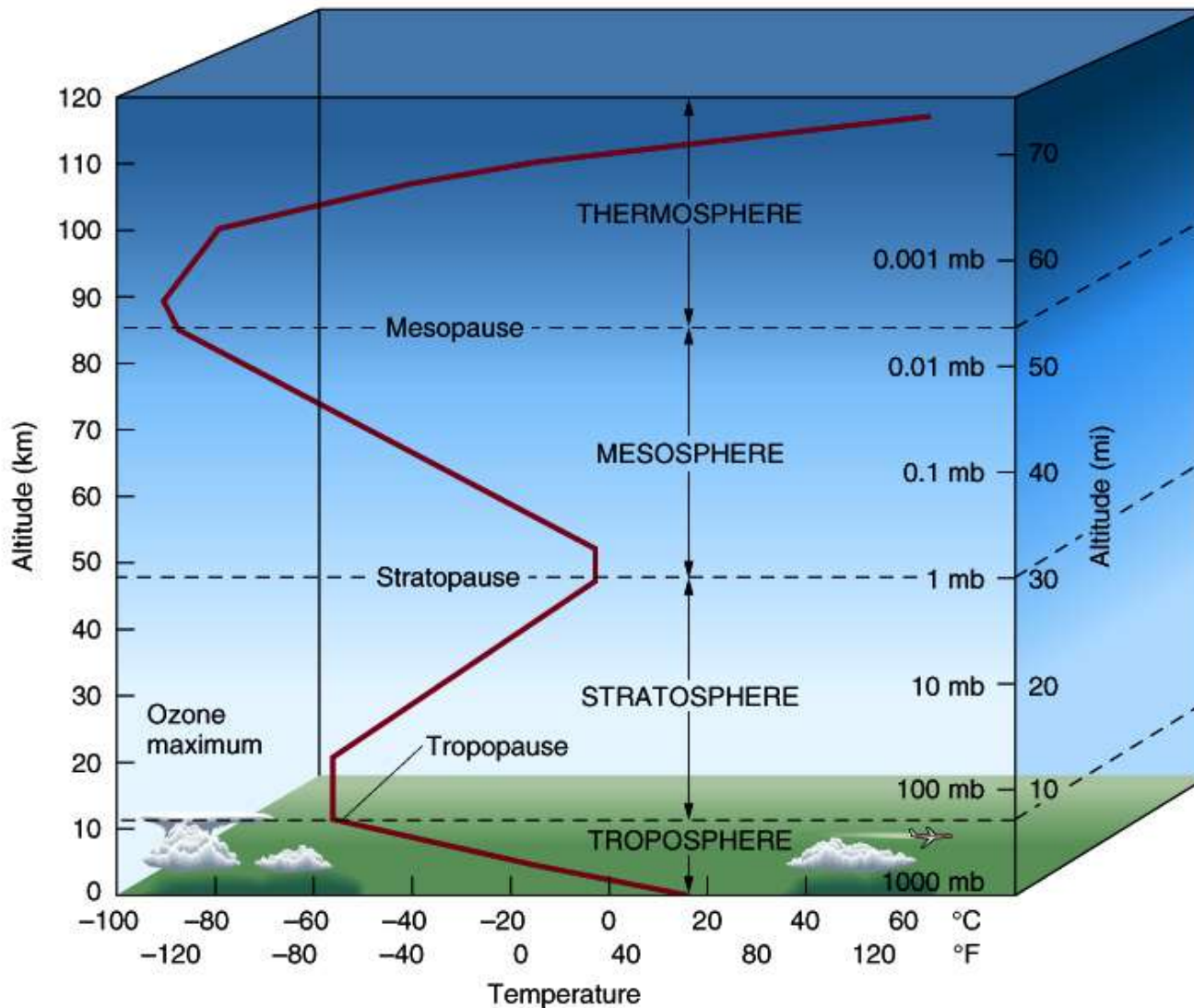


THE OXIDATION OF THE ATMOSPHERE ALLOWED THE FORMATION OF THE STRATOSPHERIC OZONE LAYER

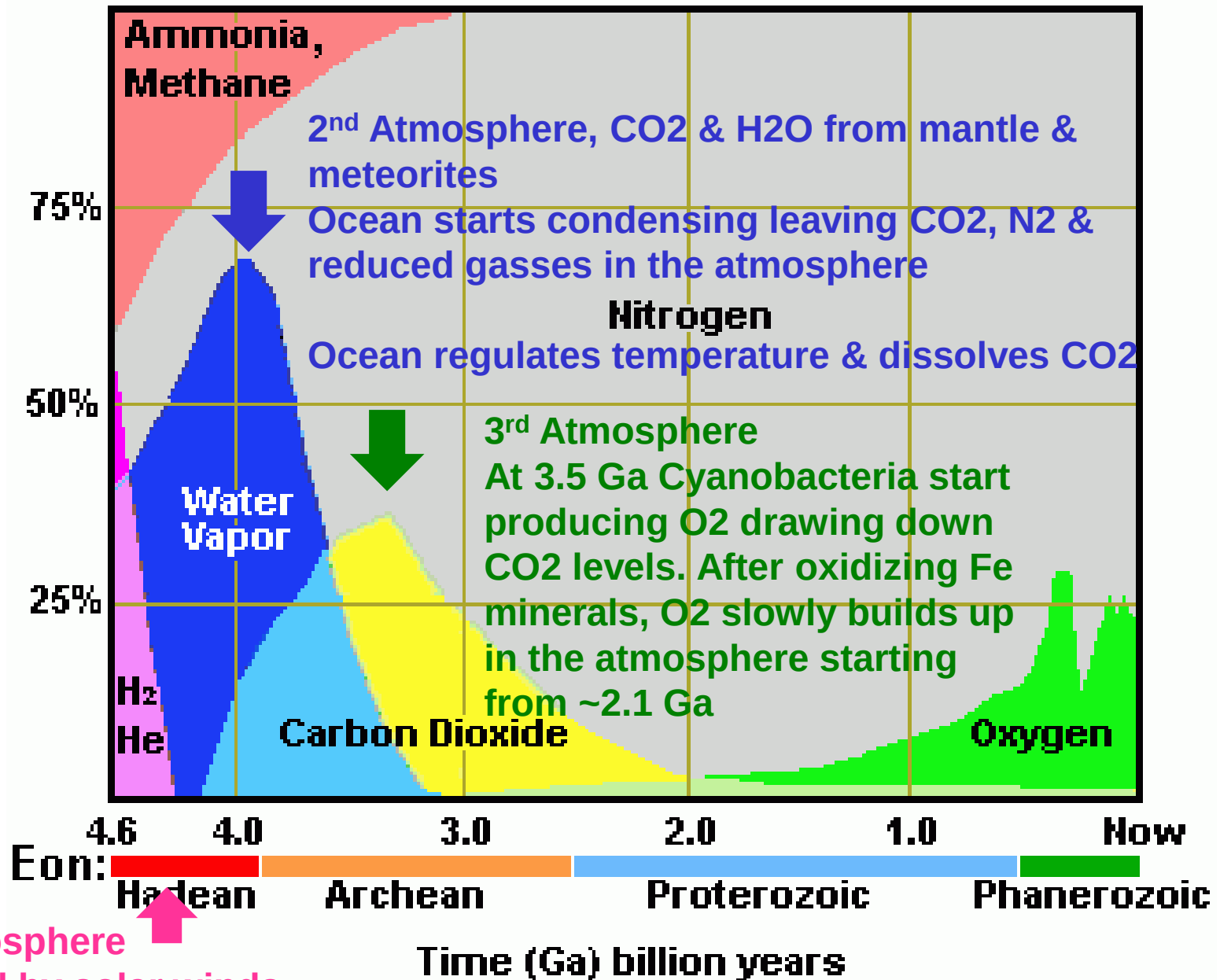


THE STRATOSPHERIC OZONE LAYER PROTECTS LIFE ON LAND FROM HARMFUL UV RAYS

THE ATMOSPHERE TODAY



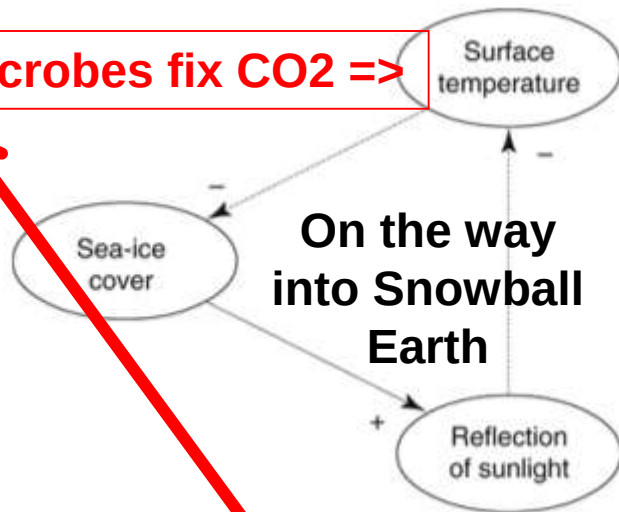
% of Atmosphere Composition of Earth's atmosphere



3 STAGES OF TEMPERATURE STABILIZATION

- 1. After the solidification of the crust the water vapour in the atmosphere condensed to form the ocean. The high heat capacity of water and the hydrological cycle helped to stabilize the temperature.**
- 2. Once photosynthetic organisms in the ocean evolved, they fixed atmospheric CO₂ into sedimentary rocks called carbonates (CaCO₃). These were buried in sedimentary basins and sometimes moved back into the mantle by plate tectonics. The lithosphere via the mechanisms weathering (which supplies Ca²⁺ ions) and volcanism became a main modulator of climate. Carbon removal triggered several “snowball earth events”**
- 3. Once life on land evolved, biological carbon contributed to climate stabilization, by converting CO₂ into coal, oil and gas deposits. However, vegetation changed land albedo and glaciations remained limited to the polar regions and mid latitudes thereafter.**

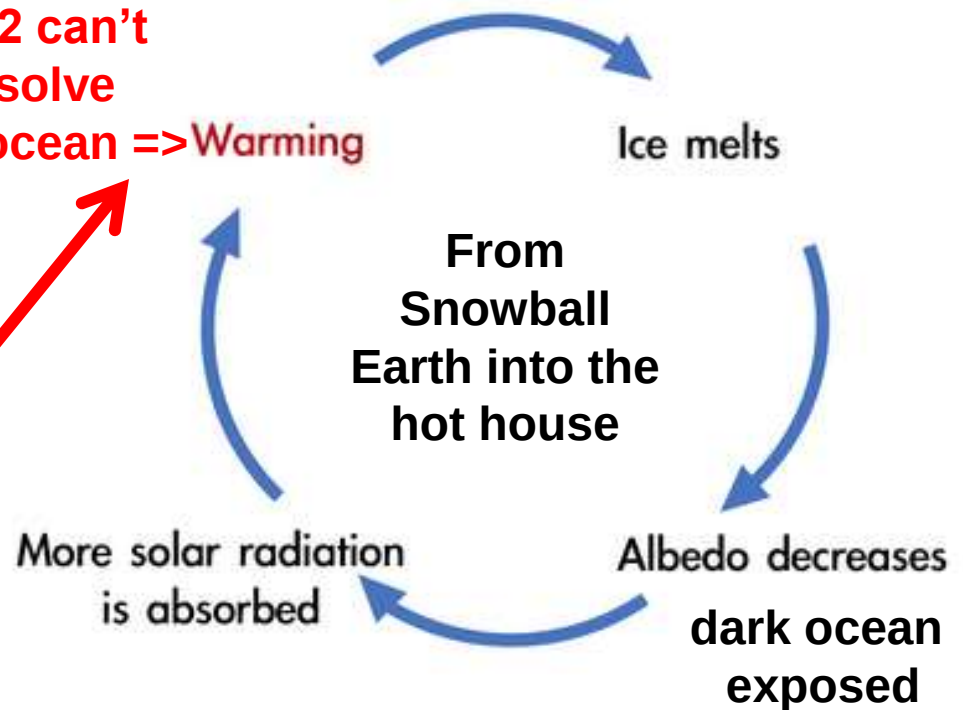
Microbes fix CO₂ =>



SNOWBALL EARTH

CO₂ can't dissolve

in ocean => Warming



Before the Cambrian explosion several such cycles happened

The big freeze: did rapid growth of the ice caps envelop the entire planet?

The nicest explanation of this is in: Earth system science a very short introduction